Exp. (1): Hooke's Law Experiment

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1. Introduction

Hooke's Law is a fundamental principle in engineering and physics that describes the relationship between the applied force and the resulting deformation in an elastic material. Understanding this relationship is crucial for structural analysis, material science, and mechanical design.

2. Objectives

The primary objectives of this experiment are:

1. To verify Hooke's Law for a helical spring.
2. To determine the spring constant (k) of the given spring.
3. To practice data collection, graphical analysis, and error estimation in an engineering context.

3. Theory

Hooke's Law states that the **restoring force** exerted by an elastic object is directly proportional to its displacement from equilibrium provided the elastic limit is not exceeded. Mathematically, it is expressed as:

$$\boldsymbol{F_{Restoring}} = -k\,\boldsymbol{x}$$

Where:

- **F** is the restoring force exerted by the spring in Newton (N).
- **k** is the spring constant (stiffness) in Newton per meter (N/m).

- **x** is the extension (change in length) in meters (m).
- The **negative sign** indicates that the restoring force always acts in the opposite direction to the displacement (x). If the spring is stretched (positive x), the restoring force pulls it back (negative F). If the spring is compressed (negative x), the restoring force pushes it outwards (positive F). This is a crucial aspect of Hooke's Law, signifying that the spring always tries to return to its equilibrium position.

In this experiment, the **applied force** (F) is the weight of the masses (m) under gravity (g = 9.8 m/s^2) that causes the spring to extend

$$F_{Applied} = mg$$

When the spring is at equilibrium (not oscillating), the magnitude of the applied force is equal to the magnitude of the restoring force; however, they act in opposite directions. Therefore, for equilibrium, we consider the magnitudes:

$$F_{Applied} = F_{Restoring}$$
$$mg = -kx$$
$$x = -\frac{g}{k}m$$

The extension (displacement) x is calculated as:

$$x = L - L_o$$

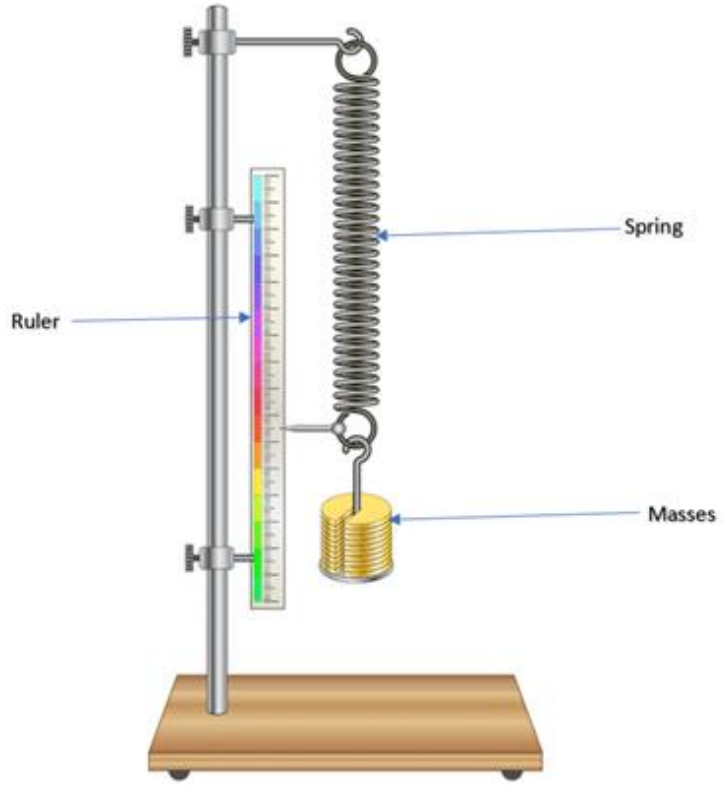
Where L is the stretched length and L_o is the original (unloaded) length.

4. Apparatus

- Retort stand and clamp
- Helical spring
- Slotted weights (e.g., 10g or 20g increments)
- Weight hanger
- Meter rule (1mm resolution)
- Pointer (attached to the spring)

5. Experimental Setup

Below is a schematic representation of the experimental arrangement:



6. Procedure

1. Setup: Secure the retort stand on a level surface. Attach the spring to the clamp and ensure it hangs freely.

2. Initial Reading: Attach the empty weight hanger to the spring. Adjust the meter rule so that the pointer aligns with a convenient mark (e.g., 0 cm or 10 cm). This is your reference point (L_o).

3. Loading:

- Add a 10 g mass to the hanger.
- Wait for the spring to stop oscillating.
- Record the new position of the pointer (L).
- Repeat this process by adding 10g increments until you have at least 6-8 data points.

5. Safety: Do not exceed the maximum load specified for the spring to avoid permanent damage or injury.

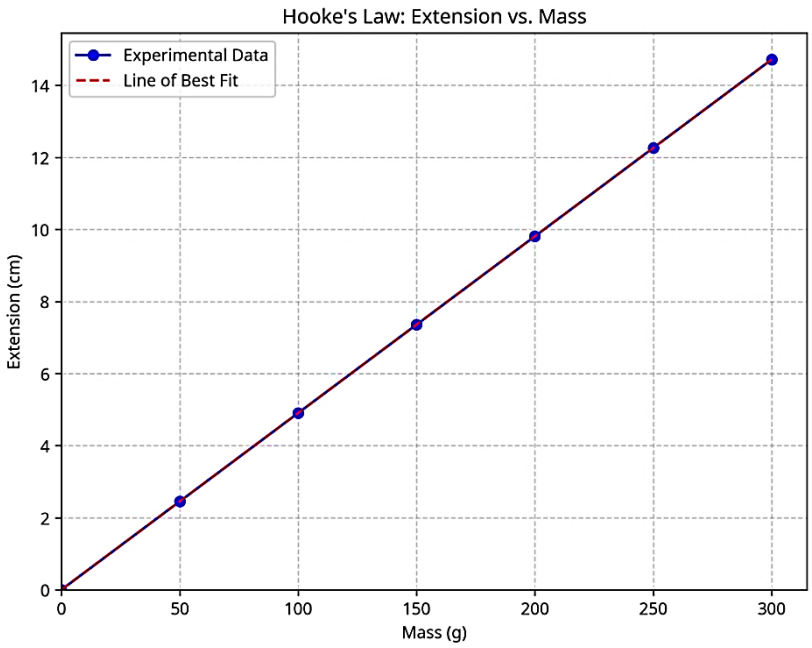
7. Data Collection

Record your observations in the following table:

Mass added (g)	Loading Position L (cm)	Extension $x = L - L_o$ (cm)
20		
30		
40		
50		
60		
70		

$$Slope = \frac{\Delta x}{\Delta m} \ (cm/g)$$

$$K = \frac{g}{slope} \ (dyne/cm)$$



8. Analysis and Calculations

1. Graphing: Plot a graph of **Extension (x)** on the y-axis versus **mass (m)** on the x-axis.

2. Linear Fit: Identify the linear region of the graph. Draw a line of best fit through the origin.

3. Spring Constant: Calculate the gradient (slope) of the linear portion of the graph.

4. Elastic Limit: Note the point where the graph starts to curve away from the straight line. This indicates the elastic limit.

9. Discussion and Error Analysis

- **Parallax Error**: Ensure your eye is level with the pointer when reading the meter rule.
- **Zero Error**: Check if the pointer is correctly aligned at the start.
- **Accuracy**: Discuss the precision of the meter rule and the weights used.

10. Engineering Application

In engineering, Hooke's Law is the basis for understanding the stress-strain relationship in materials. While this experiment uses a simple spring, the same principle applies to the elastic deformation of steel beams, bridge cables, and automotive suspension systems. The spring constant k is analogous to the stiffness of a structural component.